HEART ATTACK RISK PREDICTION USING MACHINE LEARNING

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PROJECT OVERVIEW

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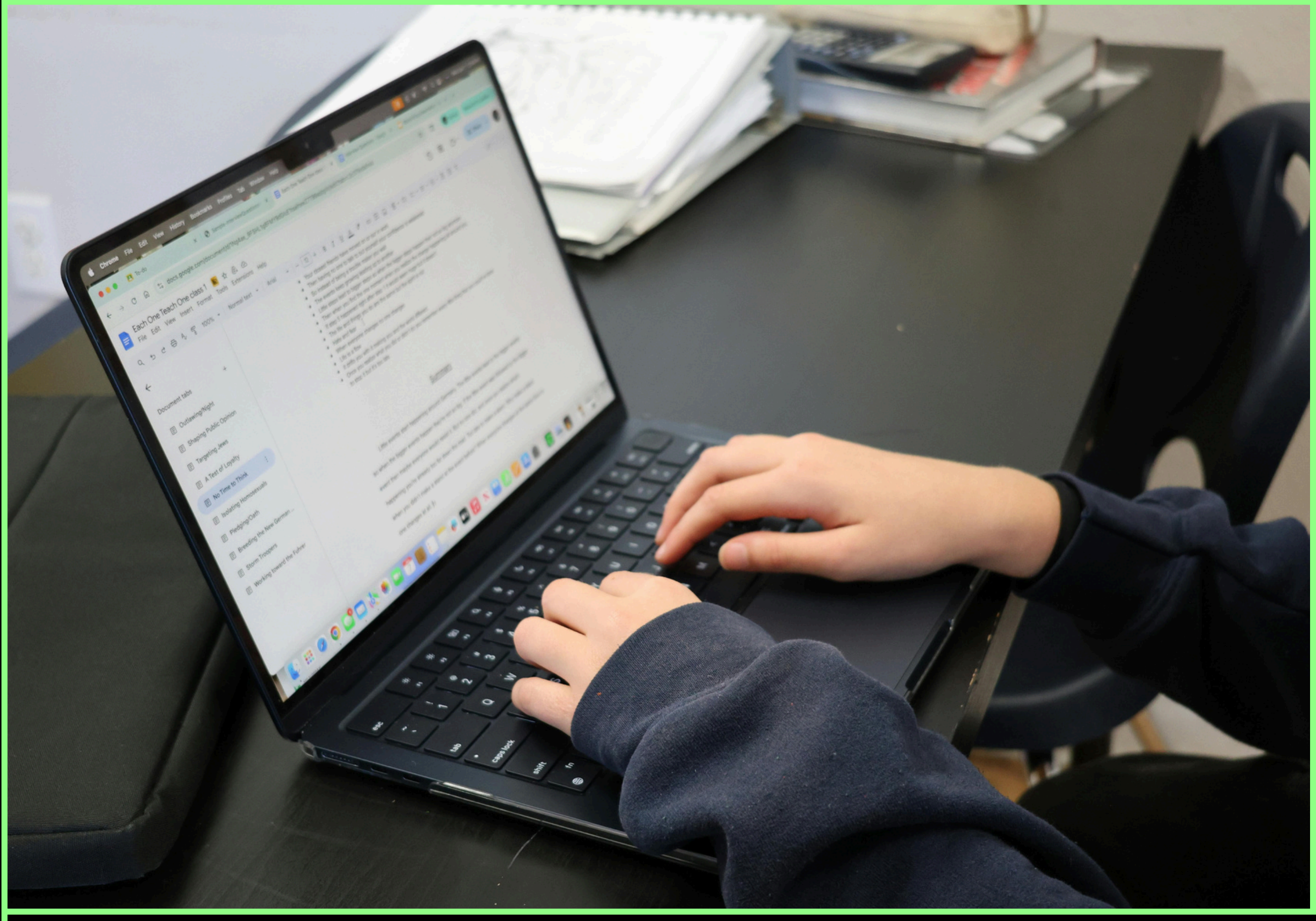
Cardiovascular diseases are one of the leading causes of death worldwide. Early prediction of heart attack risk can significantly improve prevention and treatment outcomes.

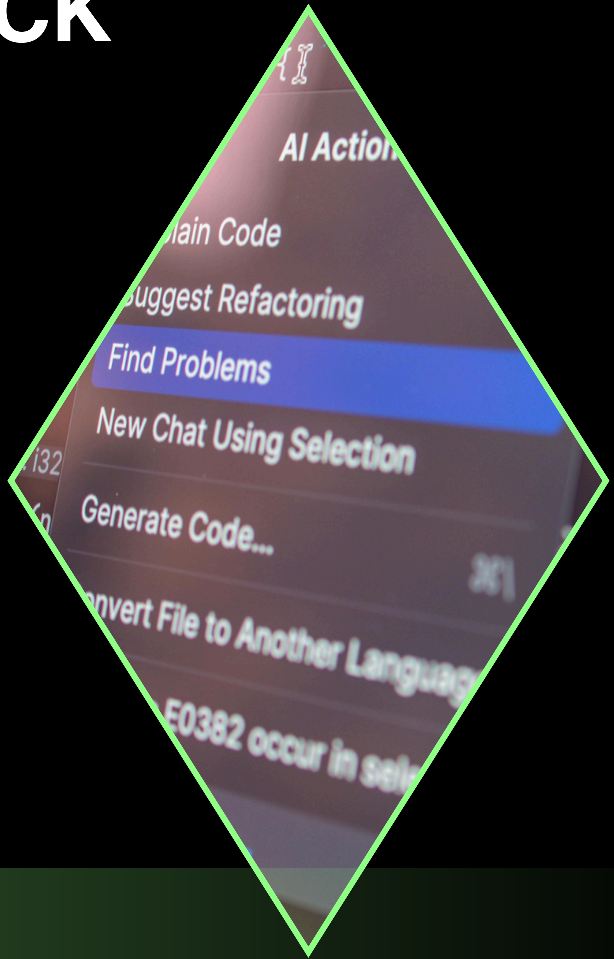
This project applies machine learning techniques to predict the risk of heart attack based on clinical and demographic data. The workflow includes data preprocessing, exploratory data analysis, feature engineering, model building, evaluation, and deployment.

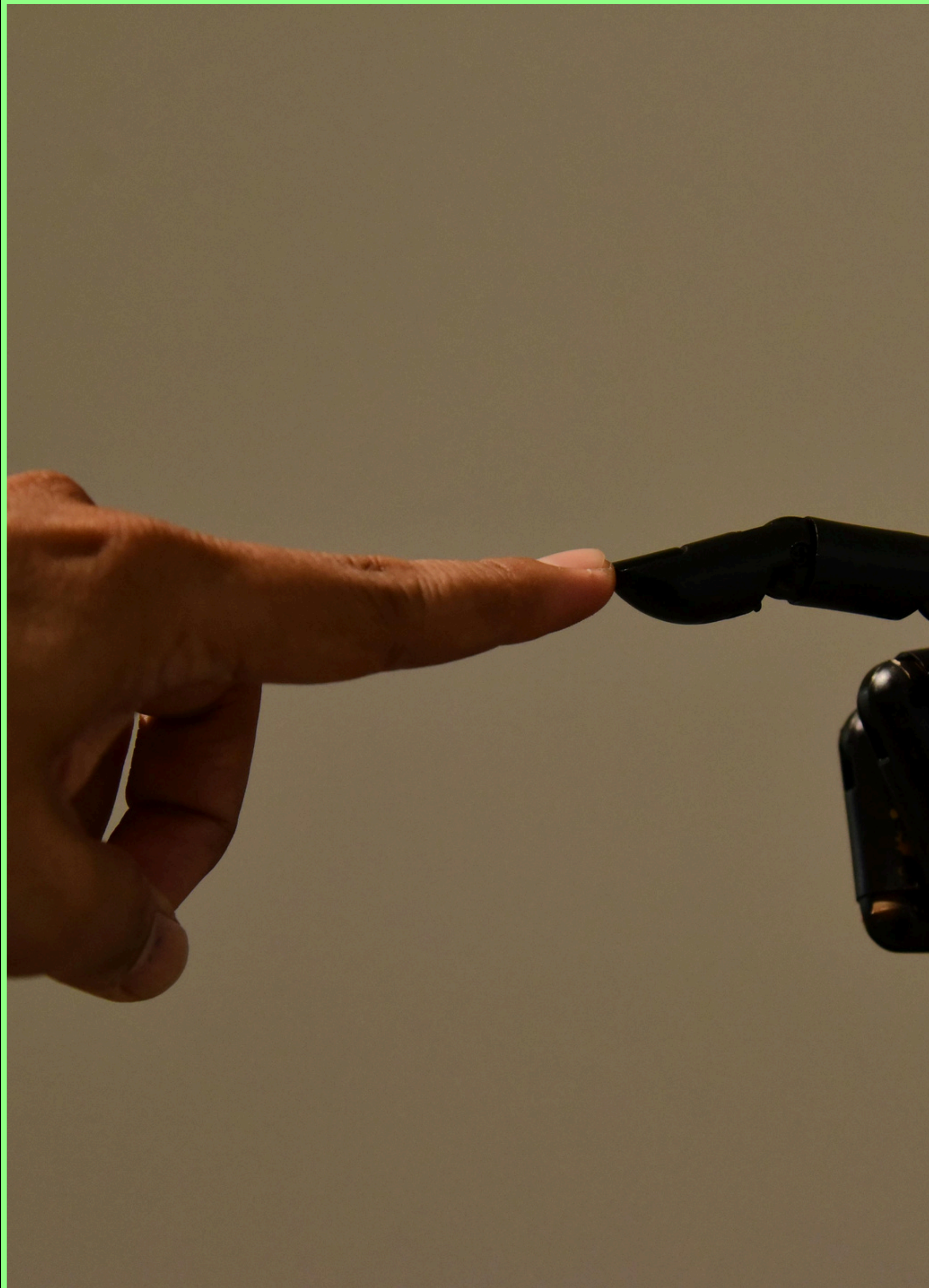


PROBLEM STATEMENT

- THE MAIN OBJECTIVE OF THIS PROJECT IS TO DEVELOP A MACHINE LEARNING MODEL CAPABLE OF PREDICTING WHETHER A PATIENT IS AT HIGH OR LOW RISK OF A HEART ATTACK BASED ON MEDICAL FEATURES.

- THE SYSTEM AIMS TO SUPPORT EARLY DIAGNOSIS AND ASSIST HEALTHCARE PROFESSIONALS IN DECISION-MAKING.





DATASET DESCRIPTION

DATASET NAME:

HEART ATTACK PREDICTION DATASET

SOURCE: KAGGLE

DATASET SIZE:

- LARGE DATASET WITH THOUSANDS OF PATIENT RECORDS
- MORE THAN 10 FEATURES

DATASET CHARACTERISTICS:

- CONTAINS NUMERICAL AND CATEGORICAL VARIABLES
- INCLUDES REAL-WORLD MEDICAL PARAMETERS
- REQUIRES PREPROCESSING AND CLEANING

TARGET VARIABLE:

HEART ATTACK RISK (0 = LOW RISK, 1 = HIGH RISK)

FEATURES USED

THE DATASET INCLUDES IMPORTANT MEDICAL FEATURES SUCH AS:

- AGE
- CHOLESTEROL LEVEL
- BLOOD PRESSURE
- HEART RATE
- SMOKING STATUS
- DIABETES STATUS
- OBESITY INDICATOR

THESE FEATURES ARE CLINICALLY RELEVANT FOR PREDICTING CARDIOVASCULAR RISK.

DATA CLEANING

DATA PREPROCESSING STEPS INCLUDED:

- HANDLING MISSING VALUES
- REMOVING DUPLICATE RECORDS
- ENCODING CATEGORICAL VARIABLES
- CONVERTING TEXT DATA INTO NUMERICAL FORMAT
- SCALING NUMERICAL FEATURES

THESE STEPS ENSURED HIGH DATA QUALITY FOR MODEL TRAINING.

EXPLORATORY DATA ANALYSIS (EDA)

EXPLORATORY DATA ANALYSIS WAS CONDUCTED TO UNDERSTAND DATA PATTERNS AND RELATIONSHIPS.

THE ANALYSIS INCLUDED:

- UNIVARIATE ANALYSIS
- BIVARIATE ANALYSIS
- CORRELATION ANALYSIS

VISUALIZATIONS USED:

- HISTOGRAMS
- BOXPLOTS
- COUNT PLOTS
- SCATTER PLOTS
- HEATMAPS

KEY INSIGHTS

THE ANALYSIS REVEALED SEVERAL IMPORTANT INSIGHTS:

- HIGHER CHOLESTEROL LEVELS ARE STRONGLY ASSOCIATED WITH INCREASED RISK.
- OLDER AGE GROUPS SHOW HIGHER PROBABILITY OF HEART ATTACK.
- HIGH BLOOD PRESSURE SIGNIFICANTLY INCREASES RISK.
- LIFESTYLE FACTORS SUCH AS SMOKING AND OBESITY CONTRIBUTE TO RISK.

FUTURE OF ENGINEERING

FEATURE ENGINEERING WAS APPLIED TO IMPROVE MODEL PERFORMANCE.

NEW FEATURES CREATED:

- AGE GROUPS (YOUNG, ADULT, SENIOR, ELDER)
- CHOLESTEROL CATEGORIES (NORMAL, BORDERLINE, HIGH)

THESE TRANSFORMATIONS HELPED CAPTURE NON-LINEAR RELATIONSHIPS IN THE DATA.

FEATURE SELECTION

FEATURE IMPORTANCE ANALYSIS WAS USED TO SELECT THE MOST RELEVANT VARIABLES.

MOST IMPORTANT FEATURES INCLUDED:

- CHOLESTEROL
- BLOOD PRESSURE
- AGE
- HEART RATE

FEATURE SELECTION HELPED IMPROVE MODEL EFFICIENCY AND REDUCE NOISE.

MACHINE LEARNING MODELS

THREE MACHINE LEARNING MODELS WERE IMPLEMENTED AND COMPARED:

- LOGISTIC REGRESSION
- RANDOM FOREST CLASSIFIER
- GRADIENT BOOSTING CLASSIFIER

THE BEST PERFORMING MODEL WAS SELECTED BASED ON EVALUATION METRICS.

MODEL TUNING

HYPERPARAMETER TUNING WAS PERFORMED USING GRIDSEARCHCV.

PARAMETERS SUCH AS:

- NUMBER OF ESTIMATORS
- MAXIMUM DEPTH

WERE OPTIMIZED TO IMPROVE MODEL PERFORMANCE.

THIS PROCESS HELPS IMPROVE ACCURACY AND GENERALIZATION.

EVALUATION METRICS

THE MODELS WERE EVALUATED USING:

- ACCURACY
- PRECISION
- RECALL
- F1-SCORE
- CONFUSION MATRIX

THESE METRICS ENSURE RELIABLE PERFORMANCE EVALUATION, ESPECIALLY FOR MEDICAL PREDICTION TASKS.

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BEST MODEL RESULTS

THE RANDOM FOREST MODEL ACHIEVED THE BEST PERFORMANCE AMONG ALL TESTED MODELS.

IT SHOWED:

- HIGH ACCURACY
- BALANCED PRECISION AND RECALL
- STRONG GENERALIZATION ABILITY

THEREFORE, IT WAS SELECTED AS THE FINAL MODEL

DEPLOYMENT

THE FINAL MODEL WAS DEPLOYED USING STREAMLIT AS AN INTERACTIVE WEB APPLICATION.

THE SYSTEM ALLOWS USERS TO:

- INPUT PATIENT HEALTH DATA
- RECEIVE REAL-TIME PREDICTION
- GET RISK CLASSIFICATION (HIGH / LOW)

THIS MAKES THE MODEL PRACTICAL FOR REAL-WORLD USAGE.

CHALLENGES

SEVERAL CHALLENGES WERE FACED DURING THE PROJECT:

- HANDLING CATEGORICAL VARIABLES
- FEATURE SCALING ISSUES
- MODEL TUNING COMPLEXITY
- DATA PREPROCESSING REQUIREMENTS

THESE CHALLENGES WERE SUCCESSFULLY ADDRESSED DURING IMPLEMENTATION.

CONCLUSION

THIS PROJECT DEMONSTRATES THE EFFECTIVENESS OF MACHINE LEARNING IN PREDICTING HEART ATTACK RISK USING CLINICAL DATA.

THE DEVELOPED SYSTEM PROVIDES ACCURATE PREDICTIONS AND CAN SUPPORT EARLY MEDICAL DECISION-MAKING.

FUTURE WORK MAY INCLUDE:

- DEEP LEARNING MODELS
- LARGER DATASETS
- CLOUD DEPLOYMENT

THANK YOU